

Motor Imagery Hand Movements Recognition Using SVM Classifier and Genetic Algorithm Optimization

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Abstract—In this paper, we introduce two types electroencephalography (EEG) features to design Brain-Computer Interfaces (BCI) applications: the first type is based on Mel Frequency Cepstral Coefficients (MFCC), and the second one is based on the wavelet transform (WT). The goal is to identify, the motor imagery (MI) of left and right hand movements using the proposed EEG signals features from 3 subjects. The features are fed to the Support Vector Machine (SVM) classifier. Evaluation of the algorithm has been done using a 10-fold cross-validation procedure. To improve the classification performance, suitable features were selected by genetic algorithm (GA) optimization. Results show the effectiveness of the combination of these features to improve the classification of MI tasks. The recognition accuracy was improved with respect to existing works using the same database and reached 90.02%.

Index Terms— brain computer interface, electroencephalogram, motor imagery, features extraction, SMV classifier, genetic algorithm, optimization.

I. INTRODUCTION

Brain-computer interface (BCI) aims to establish a relationship between the brain and the people with special needs in order to improve the quality of their life [1]. It is a system that translates all aspects of brain activities: thoughts, behaviors, and emotions generally measured using the electroencephalogram (EEG) into commands [2], [3]. The electroencephalogram is the representation of the electrical activity occurring at the surface of the brain. The main source of the EEG signals is the synchronous activity of thousands of cortical neurons.

Motor imagery (MI) is one of the types of the EEG converted into control signals for noninvasive BCI systems [4], [5]. Numerous techniques have been applied for MI tasks evaluation in a BCI system, but EEG based methods have proven to be the best because EEG signals provide a non-invasive way of estimating MI tasks. MI EEG signal is generated when a subject imagines the movement of a limb (e.g., hand or feet). The interpretation of such a brain activity is a challenge of the EEG-based BCI system. Recently, machine learning approaches have proven to be reliable for the recognition of different MI activities, where feature extraction is carried out efficiently to describe the relevant information embedded in the EEG signals succeeded by classification [6], [7]. Several methods are widely used to

extract features [8]–[10]. All existing feature selection and feature extraction methods aim to remove redundant and inappropriate features in order classification of the new features set will provide more best accuracy. Consequently, the efficiency of a BCI system, depends basically on the choice of appropriate features and classifiers [10].

Usually used feature extraction algorithms comprise wavelet transform (WT) [11], [12]. Wavelet transform has been found to be an effective tool for the time-frequency analysis of non-stationary and quasi-stationary signals. A novel method using the local binary pattern (LBP) based on the wavelet transform (WT) was proposed to characterize the behavior of EEG activities [1]. A previous study proposed a new approach for the classification to distinct the right hand and right foot MI tasks based on EEG signals using the flexible analytic wavelet transform technique [13]. The MFCC coefficients has been used for speech recognition applications and is known to provide a robust characterization of the temporal and spectral properties of the signal. Furthermore, it is considered as a popular analysis technique in many signal processing applications including EEG classification [14], [15].

Most current electroencephalography EEG-based BCIs are based on machine learning algorithms. A wide variety of classifier types are used in this field. Support vector machine (SVM) with different kernel functions, linear discriminant analysis (LDA), k-nearest neighbor (KNN), multilayer perceptron (MLP), hidden Markov model (HMM) and Fuzzy classifiers have been extensively used to identify different MI tasks [5], [7], [8], [16]–[19].

In this work, we attempt to classify MI left and right hand movement with SVM classifier by combining features based on MFCC coefficients and wavelet transform. To improve the classification performance, suitable features were selected by optimization using Genetic Algorithm (GA).

II. METHODOLOGY

A. BCI system

A major challenge of the BCI is to interpret movement intention from brain activity. Fig. 1 shows a block diagram for a BCI system. The EEG signals detected from the scalp are processed in such way to identify the activity of the subject intention. After having extracted features, the identification is

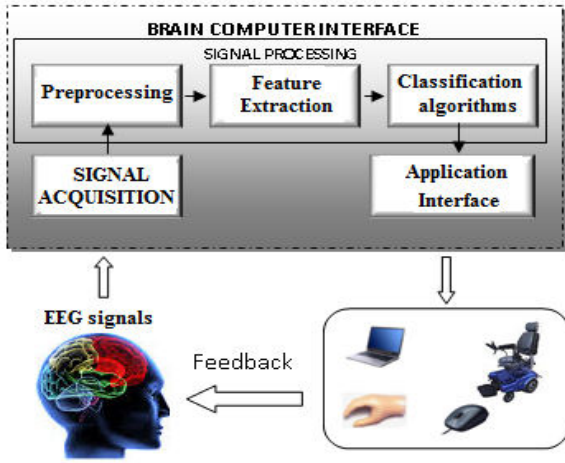


Fig. 1. Block diagram for a BCI system.

done by a trained classifier. Due to its robustness, SVM classifier was adopted in this work. Once the action is identified, an electronic interface can generate commands for specific application. The BCI technology is nothing but the machine command by thought via the EEG signals.

B. EEG data set description

The data used in this work correspond to the EEG data set IIIB of the BCI competition III [20]. Briefly, this data set contains EEG signals recorded from three subjects (O3, S4 X11) who were asked to imagine left or right hand movements when they heard a beep sound. The experiment consists of three sessions for each subject. Each session consists of four to nine runs. The recordings were made with a bipolar EEG amplifier from g.tec. The EEG signals were sampled with 125 Hz and were filtered between 0.5-30 Hz with Notch filter. A visual cue was presented to the subject in order to provide him feedback according to his imagined action. The order of left and right cues was random. Although the use of the beep sound to synchronize actions in different trails, EEG signals issue from different subjects could not be synchronized even for the same subject. Furthermore, EEG signals are of non-linearity and non-stationary nature, which makes the brain task identification very difficult. In this study, the subjects O3, S4 and X11 are named s1, s2 and s3 respectively.

C. Features extraction

Features extraction phase is the most critical operation in any classification problem. A special attention should be paid to the choice of the features. Besides to reduction utility of features, features should be information representative in a way to distinguish between classes. It is known, in pattern recognition, that the features number should be reduced to speed up the recognition process. This is usually done by dimension reduction (eg. Principle Component Analysis PCA) of features selection by optimization. The main objective of the feature selection is to select a subset from large numbers of extracted features so that distinction between classes would be strong. Fig. 2 shows the block diagram of MI left and right hand actions classification based on EEG signals. Features are extracted from SC3 and SC4 signals of the channels c3 and c4 respectively. A trained SVM classifier (using labeled data) is fed by features to predict the hand movement actions whether is it a left or right hand. The BCI system will used this task identification to interface electronically actioners to perform desired actions.

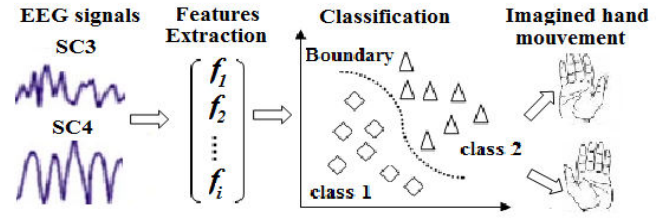


Fig. 2. Block diagram of the MI left and right hand movements classification.

1) *Wavelet transform*: Due to the non-stationary EEG signals, it is important to look for features that handle the aspect of time-frequency domain. Thus, the use of the wavelet transform becomes a logical alternative. Wavelet analysis plays a vital role in the signal processing field and is usually applied in two ways:

a) *Continuous Wavelet Transform (CWT)*: uses a wavelet function $\psi(t)$ and produces a scalogram, similar to a spectrogram for time-frequency analysis. The (CWT) of a signal $x(t)$ is given by:

$$C_x(a, b) = \frac{1}{a} \int_{-\infty}^{+\infty} x(t) \Psi\left(\frac{t-b}{a}\right) dt \quad (1)$$

where a is a positive scale parameter and b is a parameter for the position of the mother wavelet Ψ in time. The proposed features that based on the CWT use two statistical indicators, the mean and the variance as expressed in the following equations:

$$f_i = \text{var}(cw_3(i, :)) - \text{var}(cw_4(i, :)) \quad i = 1 \dots 24 \quad (2)$$

$$f_{i+24} = \text{var}(cw_3(i, :)) / \text{var}(cw_4(i, :)) \quad i = 1 \dots 24 \quad (3)$$

$$f_{i+48} = \text{mean}(cw_3(i, :)) / \text{var}(cw_4(i, :)) \quad i = 1 \dots 24 \quad (4)$$

where cw_3 and cw_4 are continuous wavelet transform of signals SC3 and SC4 respectively using the mother wavelet 'Morlet'. The 'Morlet' mother wavelet is chosen in this study according to the results found in our previous work [19]. Hence, 72 features are obtained from continuous wavelet transform.

b) *Discrete Wavelet Transform. (DWT)*: has excellent spatio-frequency localization properties, and analyzes the signal at different frequency bands with different resolutions. Associated with the low-pass $L[n]$ and high-pass $H[n]$ band filters, DWT decomposes the signal into different frequency bands with a down-sampling rate of 2 called as sub bands. The low pass filter gives the approximation coefficients (A1), while the high-pass filter gives the detail coefficients (D1). A1 is disintegrated further and the procedure repeated until reaching a specified number of decomposition levels [21]. The coefficients describe the time evolution of the signal in the corresponding frequency bands (Fig. 3). By this way, the extracted features could contain the time information at different frequency sub-bands. Therefore, features obtained with approximation coefficients provided by the successive decomposition reaching 12 levels are given in equations (5) and (6).

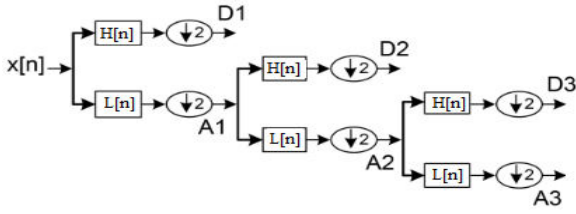


Fig. 3. DWT sub-band decomposition example.

$$f_{i+96} = \text{sum}(A_i^2) - \text{sum}(B_i^2) \quad i = 1 \dots 12 \quad (5)$$

$$f_{i+108} = \text{sum}(A_i^2) / \text{sum}(B_i^2) \quad i = 1 \dots 12 \quad (6)$$

where A_i and B_i are approximation coefficients of the decomposed signals in the DWT i^{th} levels for signals SC3 and SC4 respectively. In summary, 24 features are obtained from subtractions and ratios between energies of approximation signals resulting from DWT of SC3 and SC4 in successive decompositions.

2) *Mel-Frequency Cepstrum Coefficients (MFCC)*: Basically the MFCC are used in speech recognition as it is mentioned in the introduction. However, we tried to use these coefficients to see their effectiveness in the EEG signals classification.

Fig. 4 shows steps for calculating the MFCC coefficients. Before framing the signal, it is necessary to pass it through high pass filter to emphasis high frequencies. The filter used for this purpose is described by the following equation:

$$x(n) = s(n) - as(n-1) \quad (7)$$

Where $x(n)$ represents the output sample i.e. the filtered signal, $s(n)$ is the numerical EEG signal and a is constant number between 0.95 to 1. After having emphasis high frequencies, signals of a given segment are divided into frames of 0.75s. The frames ($x_i(n)$) are not overlapped. Then each frame is a subject of windowing process using a given window which is defined as:

$$sh_i(n) = x_i(n)w(n) \quad (8)$$

Where w is the specific window obtained after several trials and is given by:

$$w(n) = 5\text{gausswin}(M, 0.25) + 3\text{kaiser}(M), \quad 0 \leq n \leq M-1 \quad (9)$$

x_i , sh_i are the i^{th} frame and the i^{th} windowed frame respectively. M represents the length of the frame (75 ms) as a number of samples. For each new frame (after windowing), the Fourier transform is calculated to get the spectrum signal [22]. The spectrum signal is multiplied by an equidistant triangular bank filters in the Mel scale as shown in Fig. 5.

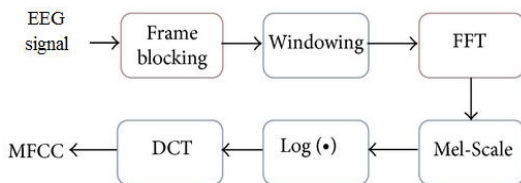


Fig. 4. Block diagram of MFCC coefficients.

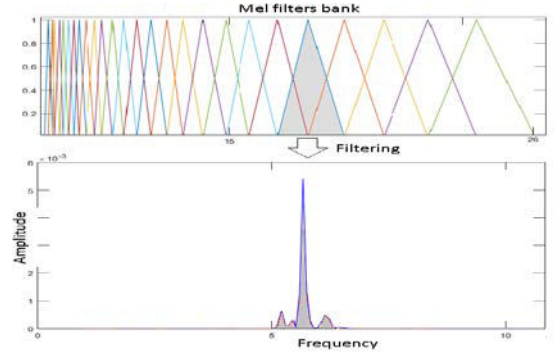


Fig. 5. Mel filters bank (up) and an example of a filtered frame in frequency domain in Hz (bottom).

Frequencies in Hertz domain and Mel domain are related by the following equation:

$$f_{mel} = a \log_{10} \left(1 + \frac{f}{b} \right), \quad a = 0.23 \text{ and } b = 3 \quad (10)$$

Where a and b are constants tuned by trial and error principle to match the EEG spectrum. Energies of filtered signals (using filters bank) are given to 'Log' operation. Logarithmic energies are transformed by the Discrete Cosine Transform (DCT) to get finally the MFCC coefficients. For each frame, we get 12 MFCC coefficients depending on the filters number. Thus, with 12 filters operating on segment of 2 s duration, 24 features are derived, with a frame duration of 0.75 s, as follows:

$$f_{mel(i+72)} = mfcc_3(i) - mfcc_4(i) \quad i = 1 \dots 24 \quad (11)$$

where $mfcc_3$ and $mfcc_4$ represent 24 MFCC coefficients for two segments of the EEG signals SC3 and SC4 respectively. Finally, a features vector of 120 features was defined as a subtractions or ratios between features from the two channels so the spatial information will be considered.

D. Machine Learning Algorithm

SVM is a well-known and powerful learning algorithm in EEG pattern recognition [5]. The general idea of SVM is to find an optimal hyperplane to maximize the margin between two classes. Maximizing the margin leads to increase the generalization capabilities and the classification accuracy [23]. If two classes are not linearly separable, kernel functions are used to transform features vector to high dimension so the problem will be linearly separable [19]. SVM is designed for binary classification and is applied in this work to classify left and right hand movements.

In this study, classification accuracies in all results are performed by 10-fold cross validation evaluation. In 10-folds cross-validation way, the data is equally divided into 10 folds, where each fold is used for testing in succession, and the remainder 9 folds are used to train the classifier. At last, the mean accuracy is obtained from the 10 folds.

III. RESULTS

Practically, subjects cannot perform tasks at the same time positions. To see the effect of time positions on the classification accuracy (CA), the proposed features were used by the linear SVM classifier to perform classification as shown in Table I.

TABLE I.

SVM CLASSIFICATION ACCURACIES (CA) IN PERCENT (%) USING ALL PROPOSED FEATURES CALCULATED FOR 2 SECONDS PROCESSING DURATION, WITH RESPECT TO TIME POSITIONS FOR ALL SUBJECTS USING 10-FOLD CROSS VALIDATION EVALUATION.

Time Position in seconds	Subjects		
	s1	s2	s3
3.00	81.23	72.72	72.96
3.25	75.89	77.17	80.74
3.50	76.46	79.43	82.22
3.75	74.46	79.84	78.88
4.00	76.56	81.45	79.07
4.25	79.62	78.76	79.44
4.50	75.30	79.73	79.44

Bold numbers indicate the best CA with respect to time positions for each subject. According to these results, it is clear that each subject performed its brain activity at a specific time position.

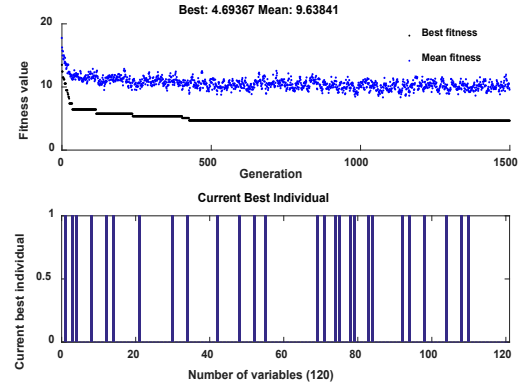
To improve the CA, the SVM classifier was optimized for each subject by GA. The concepts of GA are directly derived from natural evolution and are considered as a powerful tool for solving search and optimization problems. In this work, GA optimization was used to select suitable features so the misclassification error would be minimal.

Fig. 6 shows the GA optimization for selecting suitable features in each subject. The GA should find the best combination among 2^{120} coded binary variables cases i.e., find an appropriate set of features among the 120 features. The results indicate that the optimum CA for subject s1 is 95.31%, for subject s2 is 88.51% and for subject s3 is 86.26%.

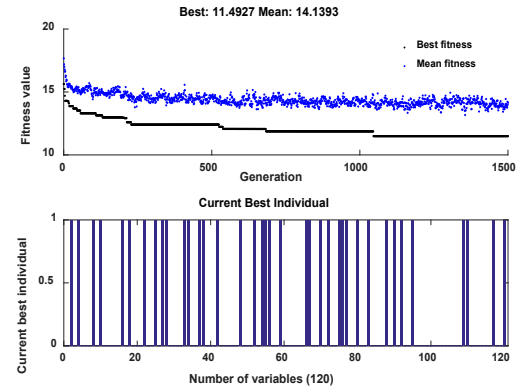
Let's now compare our results with the existing results obtained in previous works using the same database 'BCI competition IIIb'. Table II outlines the classification accuracies of different works to identify left and right hand movement based on motor imagery EEG signals. These works used different features and different classifiers. Reported results are those which gave the best performance as a mean classification value for the three studied subjects. According to Table II, selected features among 120 proposed features have improved the classification accuracy and achieved 90.02%. Note that the last result is bolded.

IV. CONCLUSION

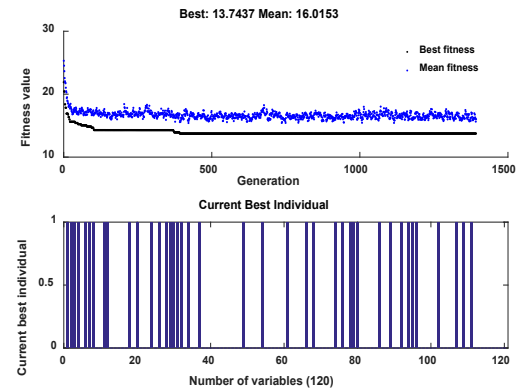
In this work, left and right hand movement in MI was identified using SVM classifier. This study reveals the difficulty of identifying brain tasks due to the nature of the EEG signals and the subject's difficulty in synchronizing with the biofeedback system.



(a)



(b)



(c)

Fig. 6. GA optimization using 10-fold cross validation evaluation for all subjects. (a) Misclassification accuracy and selected features for subject s1. (b) Misclassification accuracy and selected features for subject s2. (c) Misclassification accuracy and selected features for subject s3.

TABLE II.

SUMMARY OF AVERAGE CLASSIFICATION ACCURACIES FOR THE THREE SUBJECTS IN PREVIOUS WORKS USING THE SAME DATABASE.

Works	Features	Classifiers	s1	s2	s3	Mean value [%]
Lotte et al. [8]	Band Power	SVM	86.80	75.90	75.40	79.36
Zhong et al. [9]	Band Power	VB	89.30	72.96	75.18	78.96
Brodu et al. [16]	Wavelet features (Morlet)	LDA	—	81.50	80.40	80.95
Brodu et al. [10]	BP, Multifractal, Complexity	LDA	—	81.70	80.90	81.30
Bashashati et al. [17]	Wavelet features (Morlet)	LR	82.39	83.89	78.15	81.47
Mebarkia and Reffad [19]	16 Diversified features	SVM	91.00	89.57	88.85	89.80
This work	MFCC and Wavelet features	SVM	95.31	88.51	86.26	90.02

In the feature extraction phase, we proposed a feature vector of 120 elements derived from the wavelet transform and the MFCC coefficients. The features vector is fed into linear SVM classifier to perform the classification. To improve the accuracy, the linear SVM classifier was optimized for each subject. The optimization was performed by GA in order to select suitable features.

Usually, the MFCC coefficients are used in the applications related to the speech signal. Although most of the extracted features are based on the wavelet transform, the study shows that the features based on the MFCC coefficients improved the recognition accuracy of brain tasks based EEG signals and reached 90.02%.

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